S-F-695G
INT. AMENDMENT-3 (Army-GL)
December 30, 1987
SUPERSEDING
INT. AMENDMENT-2 (Army-GL)
March 28, 1984

FEDERAL SPECIFICATION

FRYERS, DEEP-FAT (ELECTRIC)

This interim amendment was developed by the U.S. Army Natick Research, Development, and Engineering Center (GL), Natick, MA 01760, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this interim amendment as a valid exception to Federal specification S-F-695G, dated June 25, 1981.

PAGE 3

2.1, under "Federal Specifications": Delete "QQ-S-698 - Steel, Sheet and Strip, Low-Carbon".

2.1, under "Military Standards": Add the following:

"MIL-STD-167/1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurements of.

MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities."

PAGE 4

2.2, under "American Society for Testing and Materials (ASTM)": Add "A 366 - Steel, Carbon, Cold Rolled Sheet, Commercial Quality".

PAGE 5

3.4.1, line 2: Delete "QQ-S-698" and substitute "ASTM A 366".

AMSC N/A FSC 7310

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PAGE 6

3.5, line 7, after "(submarine use)": Add "The height of the type VI fryer from the deck to the top of the removable cover shall be 36 ± 1/2 inches."

PAGE 7

3.5.2.2.1, line 4, following "fryer": Add "except that test switches which are used periodically to test the safety thermostat circuit may be located elsewhere provided they are located for safe and convenient access from the front of the fryer".

3.6.1.1, line 3, following "stainless steel": Add "or nickel plated steel".

PAGE 10

3.6.8: Delete and substitute:

"3.6.8 Thermostat. An adjustable thermostat shall be provided with a minimum range from 200° to 275°F and a maximum range from 375° to 400°F. The thermostat shall have a positive "OFF" position and be marked at 50°F intervals. A separate "ON-OFF" switch may be provided in addition to, or in lieu of, the positive "OFF" position to control power to the thermostat. Thermostat and switch shall be located so as to be readily accessible to the operator. The thermostat shall maintain an average temperature of 375 ± 5°F when tested as specified in 4.4.2.1. The temperature amplitude of the thermostat shall not exceed 25°F when tested as specified in 4.4.2.2. Instructions for calibrating the thermostat shall be provided with each fryer (see 3.8.2). The thermostat shall limit the maximum temperature to 425°F."

3.6.8.1, line 6, following "4.4.3": Delete the sentence beginning "In lieu of" and insert: "A means shall be provided to facilitate checking the operation of the over-temperature thermostat by rendering the first thermostat inoperative. A signal light shall indicate when the overtemperature thermostat is operated when tested as specified in 4.4.3."

PAGE 11

3.6.8.1 (cont'd), line 8: Delete "not exceeding 430°F." and substitute "of 430°F through 460°F.".

3.6.8.1.1: Delete first sentence and substitute: "For Naval Shipboard use (see 6.2), the deep fat fryer shall be provided with a primary thermostat shunt by pass assembly."

PAGE 12

3.6.9.3, line 2, following "above": Delete "and shall be capable of recovery to 375°F prior to completion of frying cycle".

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PAGE 14

3.9, line 2: Delete "dimensional requirements for fryers shall be as specified in figures 1, 2, and 3 and".

3.9.2: Delete and substitute:

"3.9.2 Legs and mountings. The bottom of the legs for the type I, size 1, type II and type VI fryers shall be provided with holes for the mountings. The type I fryer shall have 3/8 - 16 threaded leg mounting holes. The type II and type VI fryers shall have 9/16 inch diameter leg mounting holes for 1/2 inch diameter bolts. All legs shall be fabricated of stainless steel and the thickness shall not be less than 16 gauge. Legs shall be reinforced without cross bracing to form a rigid assembly."

PAGE 15

3.9.8, after last sentence add:

"The deep fat fryers shall be insulated such that the exterior surface temperature does not exceed 140°F. Heat-resisting hand grips shall be provided in order to insulate any handlers such that the surface temperature of the grip does not exceed 120°F."

Add the following new paragraphs:

3.9.11 U. L. requirement. The unique requirement for fryers to be connected to a remote shunt trip disconnect (see 3.6.8.1) is recognized to preclude overall U.L. approval of the complete fryer. In lieu thereof, the fryer shall be constructed of U. L. approved components and in such a manner that the fryer could be U. L. approved upon incorporation of an integral over temperature thermostat control system. When style 2 swing-up heating elements are supplied in fryers, the thermostatic controls may be mounted in the tilting head."

3.9.12 Electromagnetic compatibility. When specified for shipboard use (see 6.2), the units shall be designed and equipped for electromagnetic capability in accordance with the requirements of MIL-STD-461, class A4 for surface ships and class A5 for submarines. The equipment shall meet the emission and susceptibility requirements for CE01, CE03, and RE02 (see 4.4.11).

3.9.13 Human factors criteria (For Navy use only). Human factors engineering criteria, principles and practices, as defined in MIL-STD-1472, shall be incorporated into the design of the deep fat fryer. The equipment shall permit safe and efficient maintenance and operational tasks by the 5th percentile female to the 95th percentile male as defined in MIL-STD-1472. Sufficient clearance or free area shall be provided around an item so that an individual with applicable body dimensions and physical capabilities can safely operate,

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maintain, remove or replace that item. When establishing accessibility requirements, both physical and visual access must be provided, along with access for the use of any tools, test equipment, or repair parts. Controls or switches needed for operation or maintenance shall be selected and integrated into the machines so as to meet the applicable requirements of MIL-STD-1472. When inspecting for defects and performing tests, the equipment shall adhere to the human factors engineering considerations listed herein.

PAGE 17

Table IV, defect opposite "Signal light or lights": Delete and substitute "Not provided (power "ON" and "overtemperature thermostat" lights)".

PAGE 18

4.2.2.3, line 2, after "and": Add ", when required for shipboard use,".

PAGE 19

4.4.3: Delete the second and third sentences and substitute: "The first thermostat shall be rendered inoperative in accordance with the manufacturer's operating instructions. The hydrogenated vegetable oil temperature shall be increased until the overtemperature thermostat specified in 3.6.8.1 opens the heating element circuit."

PAGE 20

4.4.3 (cont'd), line 4, after "times": Delete the remainder of the sentence.

Line 5: Delete "sall" and substitute "shall".

4.4.4.2 (a) last sentence: Delete.

PAGE 22

4.4.9(a): Delete and substitute the following:

"(a) Shipboard vibration test. The deep fat fryer under normal operating conditions, shall be tested in accordance with MIL-STD-167/1, type I equipment. The deep fat fryer shall be secured to the test machine in the same manner that it will be secured on shipboard. Failure of the deep fat fryer to perform its function during and after testing, or to meet the requirements of 3.9.5 shall constitute failure of this test."

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Add the following new paragraph:

4.4.11 Electromagnetic interference test. When required for shipboard use, the units shall be tested by the contractor in accordance with test methods CE01, CE03 and RE02 of MIL-STD-462. The contractor shall furnish written certification that the requirement meets MIL-STD-461 requirements. Nonconformance with the requirements specified shall constitute failure of the test.

PAGE 23

6.2, (i) line 1, after "3.6.8.1": Add ", 3.6.8.1.1".

Add the following new paragraph:

"6.2.2 Technical manuals. The requirement for technical manuals should be considered when this document is cited on a contract. If technical manuals are required, a contract exhibit must be prepared to fully describe statement of work criteria and delivery instructions, and cite the applicable technical manual. The technical manuals must be acquired by separate Contract Line Item Number (CLIN) in the contract."

PAGES 24 and 25

Delete Figures 1, 2 and 3.

MILITARY INTERESTS:

Custodians

Army - GL
Navy - YD
Air Force - 99

Review Activities

Army - MD, TS
Navy - SA, MC, SH
DLA - GS

User Activity

Army - CE

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
VA - OSS

PREPARING ACTIVITY:

Army - GL

Project No. 7310-A745

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